# EXAM FORMULA SHEET

Statistical Foundation of Data Science
Quick Reference Guide

CSU1658

December 2025

## Contents

# 1 Descriptive Statistics

**Mean (Average) - HIGH PROBABILITY (HIGH PRIORITY)**

**Population Mean:**

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i = \frac{x_1 + x_2 + \cdots + x_N}{N}$$

**Sample Mean:**

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

**Weighted Mean:**

$$\bar{x}_w = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i} = \frac{w_1x_1 + w_2x_2 + \cdots + w_nx_n}{w_1 + w_2 + \cdots + w_n}$$

**Geometric Mean:**

$$\bar{x}_{\text{geo}} = \sqrt[n]{x_1 \cdot x_2 \cdot \ldots \cdot x_n} = \left(\prod_{i=1}^{n} x_i\right)^{1/n}$$

**Harmonic Mean:**

$$\bar{x}_{\text{harm}} = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}} = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}}$$

**Properties:** $\mathbb{E}[\bar{X}] = \mu$, $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$, $\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$
**When to use:** Arithmetic mean for symmetric data; Geometric for growth rates; Harmonic for rates/ratios.

**Median - MEDIUM PROBABILITY**

**Ordered dataset:** Sort data in ascending order
**For odd n:**

$$\text{Median} = x_{\left(\frac{n+1}{2}\right)}$$

**For even n:**

$$\text{Median} = \frac{x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n}{2}+1\right)}}{2}$$

**When to use:** Better than mean for skewed distributions or data with outliers. Not affected by extreme values.

**Mode - LOW PROBABILITY**

**Definition:** The value that appears most frequently in the dataset.
**Properties:** - Unimodal: One mode - Bimodal: Two modes - Multimodal: Multiple modes - No mode: All values appear with same frequency
**When to use:** Categorical data, finding most common category.

## Variance and Standard Deviation - VERY HIGH PROBABILITY (HIGH PRIORITY)

**Population Variance:**

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^2 = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

**Sample Variance (Unbiased Estimator):**

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2 = \frac{\sum(x_i-\bar{x})^2}{n-1}$$

**Computational Formula (Shortcut):**

$$s^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n}x_i^2 - \frac{(\sum_{i=1}^{n}x_i)^2}{n}\right) = \frac{n\sum x_i^2 - (\sum x_i)^2}{n(n-1)}$$

**Population Standard Deviation:**

$$\sigma = \sqrt{\sigma^2}, \quad \mathrm{SD}[X] = \sqrt{\mathrm{Var}[X]}$$

**Sample Standard Deviation:**

$$s = \sqrt{s^2}, \quad \mathrm{SE}(\bar{x}) = \frac{s}{\sqrt{n}}$$

**Coefficient of Variation (Relative Dispersion):**

$$\mathrm{CV} = \frac{s}{\bar{x}} \times 100\% = \frac{\sigma}{\mu} \times 100\%$$

**Properties of Variance:**

- $\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$ (linear transformation)
- $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y)$
- $\mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\mathrm{Cov}(X,Y)$
- If $X \perp Y$ (independent): $\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$
- $\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n}\mathrm{Var}(X_i) + 2\sum_{i<j}\mathrm{Cov}(X_i, X_j)$

**Bessel's Correction:** Division by $(n-1)$ instead of $n$ makes $s^2$ unbiased: $\mathbb{E}[s^2] = \sigma^2$.
**Interpretation:** Measures spread/dispersion around mean. Larger $\sigma^2$ = more variability.

## Standard Error (SE) - HIGH PROBABILITY (HIGH PRIORITY)

**Standard Error of Mean:**

$$\mathrm{SE}(\bar{x}) = \frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}}$$

**Standard Error of Proportion:**

$$\mathrm{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}} \approx \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

**Standard Error of Difference of Means:**

$$\mathrm{SE}(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \approx \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

**Interpretation:** Measures precision of sample statistic as estimator of population parameter. Smaller SE $\Rightarrow$ more precise estimate.

### Standard Deviation - HIGH PROBABILITY (HIGH PRIORITY)

**Population SD:**

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$

**Sample SD:**

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

**Coefficient of Variation (relative SD):**

$$CV = \frac{s}{\bar{x}} \times 100\%$$

**Interpretation:** Average distance from mean (same units as data). Smaller SD = less spread.

**Frequency:** Count of occurrences of each value

$$f_i = \text{count of } x_i$$

**Relative Frequency:**

$$RF_i = \frac{f_i}{n}$$

**Cumulative Frequency:**

$$CF_i = \sum_{j=1}^{i} f_j$$

**Cumulative Relative Frequency:**

$$CRF_i = \frac{CF_i}{n}$$

### Range IQR and Percentiles

**Range:**

$$\text{Range} = x_{\max} - x_{\min}$$

**Percentiles:** Value below which P% of data falls

$$P_k = \text{value at position } \frac{k(n+1)}{100}$$

**Quartiles:** - $Q_1$ (25th percentile): 1st quartile - $Q_2$ (50th percentile): Median - $Q_3$ (75th percentile): 3rd quartile

**Interquartile Range (IQR):**

$$IQR = Q_3 - Q_1$$

**Outlier Detection with IQR:** - Lower fence: $Q_1$ - 1.5×IQR - Upper fence: $Q_3$ + 1.5×IQR - Outliers: Values outside fences

**Quick Reference**

**Quick Tips:** - Mean is sensitive to outliers; Median is robust - SD measures spread; larger SD = more variability - Always use n-1 for sample variance (Bessel's correction) - Variance units are squared; SD has same units as data - IQR detects outliers: beyond $Q_1$-1.5×IQR or $Q_3$+1.5×IQR - Correlation measures linear association, not causation

# 2 Probability Distributions

## 2.1 PMF and PDF

**Probability Mass Function (PMF) - Discrete - HIGH PROBABILITY (HIGH PRIORITY)**

**Definition:** For discrete random variable X

$$P(X = x) = p(x)$$

**Properties:**

$$0 \leq p(x) \leq 1 \text{ for all } x$$
$$\sum_{\text{all } x} p(x) = 1$$

**Expected Value:**

$$\mathbb{E}[X] = \sum_{\text{all } x} x \cdot p(x)$$

**Variance:**

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \sum x^2 p(x) - \mu^2$$

**Probability Density Function (PDF) - Continuous - HIGH PROBABILITY (HIGH PRIORITY)**

**Definition:** For continuous random variable X

$$f(x) \geq 0 \text{ for all } x$$

**Normalization:**

$$\int_{-\infty}^{\infty} f(x)\, dx = 1$$

**Probability:**

$$P(a \leq X \leq b) = \int_a^b f(x)\, dx$$

**Expected Value:**

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

**Variance:**

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)\, dx$$

### 2.2 Bernoulli Distribution

**Bernoulli - MEDIUM PROBABILITY**

**PMF:** Single trial with success probability p

$$P(X = x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$$

Or: $P(X = x) = p^x(1-p)^{1-x}$ for x $\in$ 0,1
**Expected Value:**

$$\mathbb{E}[X] = p$$

**Variance:**

$$\text{Var}(X) = p(1-p)$$

**When to use:** Single yes/no trial, coin flip, binary outcome.

### 2.3 Binomial Distribution

**Binomial Distribution - HIGH PROBABILITY (HIGH PRIORITY)**

**PMF:** n independent trials, probability p of success

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$
**Expected Value:**

$$\mathbb{E}[X] = np$$

**Variance:**

$$\text{Var}(X) = np(1-p)$$

**Standard Deviation:**

$$\sigma = \sqrt{np(1-p)}$$

**When to use:** Fixed number of independent trials, each with same success probability. Count of successes.

### 2.4 Poisson Distribution

**Poisson Distribution - HIGH PROBABILITY (HIGH PRIORITY)**

**PMF:** Events occurring at rate $\lambda$ per unit time

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

**Expected Value:**

$$\mathbb{E}[X] = \lambda$$

**Variance:**

$$\text{Var}(X) = \lambda$$

**Standard Deviation:**

$$\sigma = \sqrt{\lambda}$$

**When to use:** Rare events, arrivals per unit time, events in fixed interval.
**Poisson Approximation to Binomial:** When n large, p small, np = $\lambda$ moderate.

## Central Limit Theorem (CLT) - MEDIUM PROBABILITY

**Statement:** For large n, the sampling distribution of the sample mean approaches normal distribution.

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

**Standardized form:**

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

**Requirements:** - Sample size n $\geq$ 30 (rule of thumb) - Independent observations - Identically distributed
**Applications:** Hypothesis testing, confidence intervals, approximations

## Law of Large Numbers (LLN)

**Weak LLN:** As n $\to \infty$, sample mean converges in probability to population mean.

$$\lim_{n\to\infty} P(|\bar{X}_n - \mu| > \epsilon) = 0$$

**Practical meaning:** Larger samples give better estimates.

## Normal Distribution $\mathcal{N}(\mu, \sigma^2)$ - VERY HIGH PROBABILITY (HIGH PRIORITY)

**Probability Density Function:**

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad x \in \mathbb{R}$$

**Standard Normal $\mathcal{N}(0, 1)$:**

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad z \in \mathbb{R}$$

**Cumulative Distribution Function:**

$$\Phi(z) = P(Z \leq z) = \int_{-\infty}^{z} \phi(t)\,dt = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}\,dt$$

**Standardization (Z-transformation):**

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1), \quad P(a \leq X \leq b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

**Parameters:** $\mathbb{E}[X] = \mu$, $\text{Var}(X) = \sigma^2$, Skewness = 0, Excess Kurtosis = 0
**Empirical Rule (68-95-99.7 Rule):**

- $P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.6827$ (68.27%)

- $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.9545$ (95.45%)
- $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.9973$ (99.73%)

**Properties:**

- Symmetric about $\mu$: $f(\mu + x) = f(\mu - x)$
- Bell-shaped, unimodal
- $\Phi(-z) = 1 - \Phi(z)$ (symmetry of CDF)
- Linear transformation: If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$
- Sum of independent normals: $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2), X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$
  $\Rightarrow X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

**Critical Values (Common):** - $z_{0.10} = 1.28$, $z_{0.05} = 1.645$, $z_{0.025} = 1.96$, $z_{0.01} = 2.33$, $z_{0.005} = 2.576$

## 2.5 Exponential Distribution

**Exponential Distribution - LOW PROBABILITY**

**PDF:** For rate parameter λ ¿ 0

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

**CDF:**

$$F(x) = 1 - e^{-\lambda x}$$

**Expected Value:**

$$\mathbb{E}[X] = \frac{1}{\lambda}$$

**Variance:**

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

**Memoryless Property:**

$$P(X > s + t | X > s) = P(X > t)$$

**When to use:** Time between events, waiting times, lifetimes

**Normal Distribution - VERY HIGH PROBABILITY (HIGH PRIORITY)**

**PDF:** X ∼ N($\mu$, $\sigma^2$)

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

**Standard Normal:** Z ∼ N(0,1)

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

**Standardization (Z-score):**

$$Z = \frac{X - \mu}{\sigma}$$

**Properties:** - Mean = Median = Mode = $\mu$ - 68% within $\mu \pm \sigma$ - 95% within $\mu \pm 1.96\sigma$ - 99.7% within $\mu \pm 3\sigma$ (Empirical Rule)

**Linear Transformation:** If X ∼ N($\mu$, $\sigma^2$), then

$$aX + b \sim N(a\mu + b, a^2\sigma^2)$$

**Sum of Normals:** If X $\sim$ N($\mu_1$, $\sigma_1{}^2$) and Y $\sim$ N($\mu_2$, $\sigma_2{}^2$) independent, then

$$X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

**Quick Reference**

**Distribution Selection Guide:** - Binary outcome $\rightarrow$ Bernoulli - Count of successes in n trials $\rightarrow$ Binomial - Rare events/arrivals $\rightarrow$ Poisson - Continuous, symmetric $\rightarrow$ Normal/Gaussian

# 3 Hypothesis Testing

## Hypothesis Testing Framework - VERY HIGH PROBABILITY (HIGH PRIORITY)

**Step 1: State Hypotheses**

$$H_0 : \text{Null hypothesis (no effect, equality)}$$
$$H_1 \text{ or } H_a : \text{Alternative hypothesis}$$

**Types of Tests:** - Two-tailed: $H_1$: $\mu \neq \mu_0$ - Right-tailed: $H_1$: $\mu$ ¿ $\mu_0$ - Left-tailed: $H_1$: $\mu$ ¡ $\mu_0$
**Step 2: Choose Significance Level**

$$\alpha = 0.05 \text{ (common), or } 0.01, 0.10$$

**Step 3: Calculate Test Statistic**
**Step 4: Find P-value or Critical Value**
**Step 5: Decision** - If p-value ¡ $\alpha$: Reject $H_0$ - If p-value $\geq \alpha$: Fail to reject $H_0$

## Z-Test (Known $\sigma$) - HIGH PROBABILITY (HIGH PRIORITY)

**Test Statistic:**

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

**Critical Values:** - Two-tailed ($\alpha$=0.05): $\pm$1.96 - One-tailed ($\alpha$=0.05): $\pm$1.645 - Two-tailed ($\alpha$=0.01): $\pm$2.576
**When to use:** Known population $\sigma$, large sample (n $\geq$ 30), or normal population.

## T-Test (Unknown $\sigma$) - HIGH PROBABILITY (HIGH PRIORITY)

**Test Statistic:**

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

**Degrees of Freedom:** df = n - 1
**When to use:** Unknown $\sigma$, small sample, approximately normal data.
**Two-Sample T-Test (Equal Variances):**

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where pooled SD: $s_p = \sqrt{\frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}}$
df = $n_1$ + $n_2$ - 2

## Type I and Type II Errors

| | $H_0$ True | $H_0$ False |
|---|---|---|
| Reject $H_0$ | Type I Error ($\alpha$) | Correct (Power) |
| Fail to Reject $H_0$ | Correct (1-$\alpha$) | Type II Error ($\beta$) |

**Type I Error:** False positive, $\alpha$ = P(Reject $H_0$ — $H_0$ true)
**Type II Error:** False negative, $\beta$ = P(Fail to reject $H_0$ — $H_0$ false)

**Power:** 1 - $\beta$ = P(Reject $H_0$ — $H_0$ false)

## Confidence Intervals

**For Mean (known $\sigma$):**

$$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = \left[\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

**For Mean (unknown $\sigma$):**

$$\bar{x} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}$$

**For Proportion:**

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

**For Difference of Means (Independent Samples):**

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2,df}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

**For Variance (Chi-Square):**

$$\left[\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}\right]$$

**Common Critical Values:** - 90% CI: $z_{0.05} = 1.645$, $t_{0.05,\infty} = 1.645$ - 95% CI: $z_{0.025} = 1.96$, $t_{0.025,30} \approx 2.042$ - 99% CI: $z_{0.005} = 2.576$, $t_{0.005,30} \approx 2.750$
**Interpretation:** $100(1-\alpha)$% of intervals constructed this way contain true parameter.

## Effect Sizes - HIGH PROBABILITY (HIGH PRIORITY)

**Cohen's d (Standardized Mean Difference):**

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}} \quad \text{where} \quad s_{\text{pooled}} = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$$

Interpretation: $|d| = 0.2$ (small), 0.5 (medium), 0.8 (large)
**Cohen's d (Single Sample):**

$$d = \frac{\bar{x} - \mu_0}{s}$$

**Hedges' g (Corrected for Small Samples):**

$$g = d \times \left(1 - \frac{3}{4(n_1+n_2)-9}\right)$$

**Eta-Squared (Effect Size for ANOVA):**

$$\eta^2 = \frac{\text{SS}_{\text{between}}}{\text{SS}_{\text{total}}} = \frac{\text{SS}_B}{\text{SS}_B + \text{SS}_W}$$

**Omega-Squared (Less Biased):**

$$\omega^2 = \frac{\text{SS}_B - (k-1)\text{MS}_W}{\text{SS}_{\text{total}} + \text{MS}_W}$$

## Quick Reference

**Quick Decision Rules:** - p-value ¡ 0.05 → Reject $H_0$ (significant result) - —Z— ¿ 1.96 → Reject $H_0$ at 5% level (two-tailed) - Confidence interval not containing $H_0$ value → Reject $H_0$

## Chi-Square Test - HIGH PROBABILITY (HIGH PRIORITY)

**Goodness of Fit Test:**

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

where: - $O_i$ = observed frequency - $E_i$ = expected frequency - k = number of categories
df = k - 1
**Test of Independence (Contingency Table):**

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

**Expected Frequency:**

$$E_{ij} = \frac{(\text{Row}_i \text{ Total}) \times (\text{Col}_j \text{ Total})}{\text{Grand Total}}$$

df = (r-1)(c-1)
**Assumption:** All expected frequencies $\geq 5$
**Effect Size - Cramér's V:**

$$V = \sqrt{\frac{\chi^2}{n \times \min(r-1, c-1)}}$$

Interpretation: 0.1 (small), 0.3 (medium), 0.5 (large)

## ANOVA (Analysis of Variance) - MEDIUM PROBABILITY

**One-Way ANOVA F-statistic:**

$$F = \frac{MSB}{MSW} = \frac{SSB/(k-1)}{SSW/(N-k)}$$

where: - k = number of groups - N = total sample size - SSB = Sum of Squares Between groups - SSW = Sum of Squares Within groups
**Sum of Squares:**

$$SST = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (x_{ij} - \bar{x})^2$$

$$SSB = \sum_{i=1}^{k} n_i (\bar{x}_i - \bar{x})^2$$

$$SSW = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$$

**Relationship:** SST = SSB + SSW
**Degrees of Freedom:** - df(Between) = k - 1 - df(Within) = N - k - df(Total) = N - 1
**Hypothesis:** - $H_0$: $\mu_1 = \mu_2 = ... = \mu_k$ - $H_1$: At least one mean differs

**Effect Size - Eta-squared:**

$$\eta^2 = \frac{SSB}{SST}$$

## Paired T-Test - MEDIUM PROBABILITY

**For paired/dependent samples:**

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

where: - $d_i =$ x$_{i1} - x_{i2} (difference for pair i) - \bar{d} =$ mean of differences - $s_d =$ standard deviation of differences - df = n - 1

**When to use:** Before-after measurements, matched pairs

## Bootstrap and Resampling - LOW PROBABILITY

**Bootstrap Method:** 1. Draw n samples with replacement from original data 2. Calculate statistic of interest (mean, median, etc.) 3. Repeat B times (typically B = 1000-10000) 4. Use distribution of bootstrap statistics for inference

**Bootstrap Confidence Interval (Percentile method):**

$$CI = [$$

$\theta^*_{\alpha/2}, \theta^*_{1-\alpha/2}]$

where $\theta^*$ are bootstrap estimates.

**When to use:** Unknown distribution, small samples, complex statistics

## Cross-Validation

**K-Fold Cross-Validation:** 1. Split data into K folds 2. Train on K-1 folds, test on remaining fold 3. Repeat K times 4. Average performance metrics

**Cross-Validation Error:**

$$CV_{(K)} = \frac{1}{K}\sum_{i=1}^{K} \text{Error}_i$$

**Leave-One-Out CV (LOOCV):** K = n (special case)

**When to use:** Model selection, hyperparameter tuning, avoiding overfitting

## Min-Max Normalization - HIGH PROBABILITY (HIGH PRIORITY)

**Formula:** Scales data to [0, 1] range

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

**General range [a, b]:**

$$x_{\text{norm}} = a + \frac{(x - x_{\min})(b - a)}{x_{\max} - x_{\min}}$$

**When to use:** Neural networks, image processing, distance-based algorithms (KNN, K-Means).

**Pros:** Preserves relationships, bounded range.

**Cons:** Sensitive to outliers.

## Z-Score Normalization (Standardization) - HIGH PROBABILITY (HIGH PRIORITY)

**Formula:**

$$z = \frac{x - \mu}{\sigma} = \frac{x - \bar{x}}{s}$$

**Properties:** - Mean = 0 - Standard deviation = 1 - Unbounded range
**When to use:** When data should have mean 0 and SD 1, linear regression, PCA, algorithms assuming normal distribution.
**Pros:** Not bounded, handles outliers better than min-max.
**Cons:** Doesn't preserve exact relationships.

## One-Hot Encoding - MEDIUM PROBABILITY

**For categorical variable with k categories:**
Create k binary columns, each representing one category.
**Example:** Color $\in$ {Red, Green, Blue}

| Original | Red | Green | Blue |
|---|---|---|---|
| Red | 1 | 0 | 0 |
| Green | 0 | 1 | 0 |
| Blue | 0 | 0 | 1 |

**Formula:**

$$x_i^{(j)} = \begin{cases} 1 & \text{if category } = j \\ 0 & \text{otherwise} \end{cases}$$

**When to use:** Nominal categorical variables with no ordering.
**Note:** For linear models, use k-1 columns to avoid multicollinearity (dummy variable trap).

## Label Encoding

**For ordinal categorical variables:**
Assign integers 0, 1, 2, ... to categories based on order.
**Example:** Size $\in$ {Small, Medium, Large} - Small $\rightarrow$ 0 - Medium $\rightarrow$ 1 - Large $\rightarrow$ 2
**When to use:** Ordinal data with natural ordering, tree-based models.

## Robust Scaling - MEDIUM PROBABILITY

**Formula:** Uses median and IQR to be robust to outliers

$$x_{\text{scaled}} = \frac{x - \text{Median}(x)}{IQR}$$

where IQR = $Q_3$ - $Q_1$
**When to use:** Data with outliers, non-normal distributions
**Advantages:** Not affected by extreme values, preserves shape

## Log Transformation

**For skewed data:**

$$x_{\text{transformed}} = \log(x)$$

or for values close to zero:

$$x_{\text{transformed}} = \log(x+1)$$

**Box-Cox Transformation:**

$$x^{(\lambda)} = \begin{cases} \frac{x^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \log(x) & \text{if } \lambda = 0 \end{cases}$$

**When to use:** Highly skewed data, making data more normal, stabilizing variance

**Least Squares Estimates:**

$$\beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

**Alternative formula for slope:**

$$\beta_1 = r \cdot \frac{s_y}{s_x}$$

where r is correlation coefficient.

**Prediction:**

$$\hat{y} = \beta_0 + \beta_1 x$$

## Multiple Linear Regression - HIGH PROBABILITY (HIGH PRIORITY)

**Model:**

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

**Matrix Form:**

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

**OLS Solution:**

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

**Prediction:**

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$$

## Regression Metrics - VERY HIGH PROBABILITY (HIGH PRIORITY)

**Residuals:**

$$e_i = y_i - \hat{y}_i$$

**Sum of Squared Errors (SSE):**

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n e_i^2$$

**Total Sum of Squares (SST):**

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

**Regression Sum of Squares (SSR):**

$$SSR = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

**Relationship:** SST = SSR + SSE
**R-squared (Coefficient of Determination):**

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

Interpretation: Proportion of variance in y explained by x.
**Adjusted R-squared:**

$$R^2_{\text{adj}} = 1 - \frac{SSE/(n-p-1)}{SST/(n-1)}$$

Penalizes adding unnecessary predictors.
**Mean Squared Error (MSE):**

$$MSE = \frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \frac{SSE}{n}$$

**Root Mean Squared Error (RMSE):**

$$RMSE = \sqrt{MSE} = \sqrt{\frac{SSE}{n}}$$

**Mean Absolute Error (MAE):**

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

## 3.1 Logistic Regression

**Logistic Regression - HIGH PROBABILITY (HIGH PRIORITY)**

**Model:** For binary outcome (0 or 1)

$$P(Y=1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p)}}$$

**Logit (Log-Odds):**

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

**Odds:**

$$\text{Odds} = \frac{p}{1-p} = e^{\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p}$$

**Sigmoid Function:**

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

**Decision Rule:**

$$\hat{y} = \begin{cases} 1 & \text{if } P(Y = 1|X) \geq 0.5 \\ 0 & \text{if } P(Y = 1|X) < 0.5 \end{cases}$$

**When to use:** Binary classification problems.

### 3.2 Polynomial Regression

**Polynomial Regression - MEDIUM PROBABILITY**

**Degree 2 (Quadratic):**

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon$$

**Degree p:**

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_p x^p + \epsilon$$

**Note:** Still linear in parameters $\beta$, so use OLS.
**When to use:** Non-linear relationships, curved patterns in data.
**Warning:** High degrees can cause overfitting.

### 3.3 Ridge Regression (L2 Regularization)

**Ridge Regression - MEDIUM PROBABILITY (HIGH PRIORITY)**

**Objective:** Minimize

$$SSE + \lambda \sum_{j=1}^{p} \beta_j^2$$

**Solution:**

$$\hat{\boldsymbol{\beta}}_{\text{ridge}} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}$$

where $\lambda \geq 0$ is regularization parameter.
**Effect:** - $\lambda = 0$: Ordinary least squares - $\lambda$ ¿ 0: Shrinks coefficients toward zero - $\lambda \to \infty$: All coefficients $\to 0$
**When to use:** Multicollinearity, many predictors, prevent overfitting.
**Note:** Standardize features before applying ridge regression.

**LASSO Regression (L1 Regularization)**

**Objective:** Minimize

$$SSE + \lambda \sum_{j=1}^{p} |\beta_j|$$

**Effect:** Can shrink some coefficients exactly to zero (feature selection).
**Comparison:** - Ridge: Shrinks but keeps all features - LASSO: Can eliminate features (sparse solutions)
**Elastic Net:** Combines L1 and L2

$$SSE + \lambda_1 \sum |\beta_j| + \lambda_2 \sum \beta_j^2$$

**Logistic Regression Cost Function**

**Log-Likelihood (Cost Function):**

$$L(\beta) = -\sum_{i=1}^{n} {}^{b}\log(p_i) + (1 - y_i)\log(1 - p_i)$$

where $p_i = \frac{1}{1+e^{-(\beta_0+\beta_1 x_i)}}$
**Optimization:** Use gradient descent or Newton-Raphson
**Odds Ratio:** For one unit increase in x

$$OR = e^{\beta_1}$$

Interpretation: Multiplicative change in odds

**Residual Analysis - MEDIUM PROBABILITY (HIGH PRIORITY)**

**Residual:**

$$e_i = y_i - \hat{y}_i$$

**Standardized Residual:**

$$r_i = \frac{e_i}{s\sqrt{1 - h_{ii}}}$$

where h$_{ii} isleverage(diagonalofhatmatrix)$
**Cook's Distance (influence):**

$$D_i = \frac{r_i^2}{p}\frac{h_{ii}}{1 - h_{ii}}$$

**Rules of thumb:** - —r$_i| > 2 : Potentialoutlier - D_i > 4/n : Influentialpoint$
**Durbin-Watson Test (autocorrelation):**

$$DW = \frac{\sum_{i=2}^{n}(e_i - e_{i-1})^2}{\sum_{i=1}^{n} e_i^2}$$

Range: $0 \leq$ DW $\leq 4$ - DW $\approx 2$: No autocorrelation - DW ¡ 2: Positive autocorrelation - DW ¿ 2: Negative autocorrelation

## 4 Decision Trees

## Entropy - HIGH PROBABILITY (HIGH PRIORITY)

**Definition:** Measure of impurity/disorder

$$H(S) = -\sum_{i=1}^{c} p_i \log_2(p_i)$$

where: - c = number of classes - $p_i = proportion of class i in set S$

**Properties:** - H = 0: Pure (all same class) - H = 1: Maximum impurity (binary, 50-50 split) - Higher entropy = more disorder

**For binary classification:**

$$H = -p \log_2(p) - (1-p)\log_2(1-p)$$

## Information Gain - HIGH PROBABILITY (HIGH PRIORITY)

**Formula:**

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v)$$

where: - S = parent set - A = attribute/feature - $S_v = subset of S where attribute A has value v$

**Interpretation:** Reduction in entropy after split on attribute A.

**Decision Rule:** Choose attribute with maximum information gain.

## Gini Index (Gini Impurity) - HIGH PROBABILITY (HIGH PRIORITY)

**Formula:**

$$Gini(S) = 1 - \sum_{i=1}^{c} p_i^2$$

**For binary classification:**

$$Gini = 1 - p^2 - (1-p)^2 = 2p(1-p)$$

**Properties:** - Gini = 0: Pure (all same class) - Gini = 0.5: Maximum impurity (binary, 50-50 split) - Lower Gini = less impurity

**Gini Gain:**

$$GiniGain(S, A) = Gini(S) - \sum_{v} \frac{|S_v|}{|S|} Gini(S_v)$$

**Note:** CART (Classification and Regression Trees) uses Gini Index.

## Quick Reference

**Entropy vs Gini:** - Both measure impurity - Entropy: Logarithmic, ranges [0, 1] for binary - Gini: Quadratic, ranges [0, 0.5] for binary - Gini is faster to compute (no logarithms) - Results usually similar in practice

## Decision Tree Pruning

**Cost-Complexity Pruning ($\alpha$ parameter):**

$$\text{Cost} = \text{Error}(T) + \alpha \times |T|$$

where: - Error(T) = misclassification error of tree T - —T— = number of terminal nodes - $\alpha$ = complexity parameter
**Goal:** Find subtree that minimizes cost
**Methods:** - Reduced Error Pruning - Cost-Complexity Pruning (weakest link) - Pessimistic Error Pruning
**When to prune:** Prevent overfitting, improve generalization

**Random Forest**

**Ensemble Method:** Combines multiple decision trees
**Algorithm:** 1. Create B bootstrap samples 2. Build tree on each sample 3. For each split, consider m random features (m ¡ p) 4. Aggregate predictions (majority vote or average)
**Out-of-Bag (OOB) Error:**

$$\text{OOB Error} = \frac{1}{n}\sum_{i=1}^{n} I(y_i \neq \hat{y}_i^{OOB})$$

**Feature Importance:** Based on decrease in impurity when feature is used for splitting
**Typical values:** m = p (classification), m = p/3 (regression)

## 5 Confusion Matrix and Classification Metrics

**Confusion Matrix - VERY HIGH PROBABILITY (HIGH PRIORITY)**

**For Binary Classification:**

| | | Predicted | |
|---|---|---|---|
| | | Positive | Negative |
| 2*Actual | Positive | TP | FN |
| | Negative | FP | TN |

where: - TP = True Positives (correctly predicted positive) - TN = True Negatives (correctly predicted negative) - FP = False Positives (Type I error) - FN = False Negatives (Type II error)

**Classification Metrics - VERY HIGH PROBABILITY (HIGH PRIORITY)**

**Accuracy:**

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

**Precision (Positive Predictive Value):**

$$Precision = \frac{TP}{TP + FP}$$

Interpretation: Of all predicted positives, what fraction are actually positive?
**Recall (Sensitivity, True Positive Rate):**

$$Recall = \frac{TP}{TP + FN}$$

Interpretation: Of all actual positives, what fraction did we predict?
**Specificity (True Negative Rate):**

$$Specificity = \frac{TN}{TN + FP}$$

**False Positive Rate:**

$$FPR = \frac{FP}{FP + TN} = 1 - Specificity$$

**F1-Score (Harmonic Mean of Precision and Recall):**

$$F1 = 2 \cdot \frac{Precision \times Recall}{Precision + Recall} = \frac{2TP}{2TP + FP + FN}$$

**F-beta Score:**

$$F_\beta = (1 + \beta^2) \cdot \frac{Precision \times Recall}{\beta^2 \cdot Precision + Recall}$$

- $\beta$ ¿ 1: Emphasizes recall - $\beta$ ¡ 1: Emphasizes precision

## Quick Reference

**Metric Selection Guide:** - Balanced classes → Accuracy - Imbalanced classes → F1-Score, Precision/Recall - Cost of FP high (spam detection) → Precision - Cost of FN high (disease screening) → Recall - Trade-off between FP and FN → F1-Score

## ROC Curve and AUC - MEDIUM PROBABILITY

**Receiver Operating Characteristic (ROC):**
Plot of True Positive Rate vs False Positive Rate at different thresholds.
**TPR (Sensitivity):**

$$TPR = \frac{TP}{TP + FN}$$

**FPR:**

$$FPR = \frac{FP}{FP + TN}$$

**Area Under Curve (AUC):** - AUC = 1.0: Perfect classifier - AUC = 0.5: Random classifier - AUC ¿ 0.7: Generally acceptable - AUC ¿ 0.8: Good model - AUC ¿ 0.9: Excellent model
**When to use:** Comparing classifier performance, threshold selection

## Matthews Correlation Coefficient (MCC)

**For imbalanced datasets:**

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

**Range:** -1 ≤ MCC ≤ 1 - MCC = 1: Perfect prediction - MCC = 0: Random prediction - MCC = -1: Total disagreement
**Advantage:** Balanced measure even for imbalanced classes

**Cohen's Kappa - LOW PROBABILITY**

**Agreement measure:**

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

where: - p$_o = observedagreement - p_e = expectedagreementbychance$

**Interpretation:** - ¡ 0: No agreement - 0.01-0.20: Slight agreement - 0.21-0.40: Fair agreement - 0.41-0.60: Moderate agreement - 0.61-0.80: Substantial agreement - 0.81-1.00: Almost perfect agreement

## 6 Naive Bayes Classifier

**Bayes' Theorem - VERY HIGH PROBABILITY (HIGH PRIORITY)**

**Formula:**

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

**For Classification:**

$$P(C_k|X) = \frac{P(X|C_k) \cdot P(C_k)}{P(X)}$$

where: - C$_k = classk - X = featurevector - P(C_k) = priorprobabilityofclassk - P(X|C_k) = likelihood - P(C_k|X) = posteriorprobability$

**Naive Bayes Classification - HIGH PROBABILITY (HIGH PRIORITY)**

**Assumption:** Features are conditionally independent given class.

$$P(C_k|x_1, x_2, \ldots, x_n) \propto P(C_k) \prod_{i=1}^{n} P(x_i|C_k)$$

**Decision Rule:** Assign to class with maximum posterior probability

$$\hat{y} = \arg\max_k P(C_k) \prod_{i=1}^{n} P(x_i|C_k)$$

**Log Form (to avoid underflow):**

$$\hat{y} = \arg\max_k \left[ \log P(C_k) + \sum_{i=1}^{n} \log P(x_i|C_k) \right]$$

**Naive Bayes for Continuous Features (Gaussian)**

**Likelihood:** Assume features follow normal distribution

$$P(x_i|C_k) = \frac{1}{\sqrt{2\pi\sigma_{k,i}^2}} \exp\left( -\frac{(x_i - \mu_{k,i})^2}{2\sigma_{k,i}^2} \right)$$

where $\mu_{k,i} and \sigma_{k,i} aremeanandSDoffeatureiinclassk.$

**Naive Bayes for Discrete Features (Multinomial)**

**Likelihood:**

$$P(x_i|C_k) = \frac{\text{count}(x_i, C_k) + \alpha}{\text{count}(C_k) + \alpha \cdot |V|}$$

where: - $\alpha$ = smoothing parameter (Laplace smoothing, typically $\alpha = 1$) - —V— = vocabulary size (number of unique values)
**Laplace Smoothing:** Prevents zero probabilities
**Add-k Smoothing:** Generalization with $\alpha$ = k

**Naive Bayes Model Selection**

**Gaussian Naive Bayes:** - Continuous features - Assumes normal distribution per class
**Multinomial Naive Bayes:** - Discrete count data - Text classification (word counts)
**Bernoulli Naive Bayes:** - Binary features (present/absent) - Document classification (word presence)
**Advantages:** - Fast training and prediction - Works well with small datasets - Handles high-dimensional data - Probabilistic output
**Disadvantages:** - Independence assumption rarely holds - Can't capture feature interactions

## 7 K-Nearest Neighbors (KNN)

**KNN Algorithm - HIGH PROBABILITY (HIGH PRIORITY)**

**Steps:** 1. Choose number of neighbors k 2. Calculate distance to all training points 3. Select k nearest neighbors 4. For classification: Majority vote 5. For regression: Average of k neighbors
**Classification:**

$$\hat{y} = \text{mode}\{y_1, y_2, \ldots, y_k\}$$

**Regression:**

$$\hat{y} = \frac{1}{k}\sum_{i=1}^{k} y_i$$

**Distance Metrics - HIGH PROBABILITY (HIGH PRIORITY)**

**Euclidean Distance (L2):**

$$d(x, x') = \sqrt{\sum_{i=1}^{n}(x_i - x'_i)^2}$$

**Manhattan Distance (L1):**

$$d(x, x') = \sum_{i=1}^{n}|x_i - x'_i|$$

**Minkowski Distance (generalized):**

$$d(x, x') = \left(\sum_{i=1}^{n}|x_i - x'_i|^p\right)^{1/p}$$

- p = 1: Manhattan - p = 2: Euclidean - p → ∞: Chebyshev (max difference)

**Cosine Similarity:**

$$\text{similarity} = \frac{\mathbf{x} \cdot \mathbf{x}'}{\|\mathbf{x}\|\|\mathbf{x}'\|} = \frac{\sum x_i x_i'}{\sqrt{\sum x_i^2}\sqrt{\sum x_i'^2}}$$

Cosine Distance: d = 1 - similarity

## Weighted KNN

**Weights inversely proportional to distance:**

$$w_i = \frac{1}{d_i}$$

**Weighted Average:**

$$\hat{y} = \frac{\sum_{i=1}^{k} w_i y_i}{\sum_{i=1}^{k} w_i}$$

**When to use:** Give more importance to closer neighbors.

## Quick Reference

**Choosing k:** - k too small: Overfitting, sensitive to noise - k too large: Underfitting, over-smoothing - Rule of thumb: k = n (n = training size) - Always use odd k for binary classification (avoid ties) - Use cross-validation to find optimal k

## KNN for Imbalanced Data

**Weighted Voting:**
Give more weight to closer neighbors and adjust for class imbalance.
**Weight:**

$$w_i = \frac{1}{d_i} \times \frac{n}{n_c}$$

where: - $d_i$ = distance to neighbor i - n = total samples - n$_c$ = $samplesinclassc$
**Distance-Weighted KNN:** Closer neighbors have more influence
**Class-Balanced KNN:** Adjusts for class distribution

## Curse of Dimensionality

**Problem:** As dimensions increase, distance becomes less meaningful.
**Effect on KNN:** - All points become roughly equidistant - Nearest neighbors aren't truly ”near” - Performance degrades
**Solutions:** - Dimensionality reduction (PCA, feature selection) - Feature engineering - Use appropriate distance metrics - Increase sample size exponentially with dimensions
**Rule of thumb:** Keep dimensions p ¡¡ n (number of samples)

# 8 K-Means Clustering

## K-Means Algorithm - HIGH PROBABILITY (HIGH PRIORITY)

**Objective:** Minimize within-cluster sum of squares (WCSS)

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} \|x - \mu_i\|^2$$

where: - k = number of clusters - C$_i = clusteri - \mu_i = centroidofclusteri$
**Algorithm Steps:** 1. Initialize k centroids randomly 2. **Assignment step:** Assign each point to nearest centroid

$$C_i = \{x : \|x - \mu_i\| \leq \|x - \mu_j\| \text{ for all } j\}$$

3. **Update step:** Recalculate centroids

$$\mu_i = \frac{1}{|C_i|} \sum_{x \in C_i} x$$

4. Repeat steps 2-3 until convergence

## Within-Cluster Sum of Squares (WCSS) - HIGH PROBABILITY (HIGH PRIORITY)

**Formula:**

$$WCSS = \sum_{i=1}^{k} \sum_{x \in C_i} \|x - \mu_i\|^2$$

**Inertia:** Another name for WCSS.
**Elbow Method:** Plot WCSS vs k, choose k at ”elbow” where rate of decrease slows.

## Silhouette Score - MEDIUM PROBABILITY

**For point i:**

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

where: - a(i) = average distance to points in same cluster - b(i) = average distance to points in nearest other cluster
**Range:** -1 ≤ s(i) ≤ 1 - s(i) ≈ 1: Well clustered - s(i) ≈ 0: On cluster boundary - s(i) ¡ 0: Wrong cluster
**Overall Silhouette Score:**

$$s = \frac{1}{n} \sum_{i=1}^{n} s(i)$$

## Quick Reference

**K-Means Tips:** - Sensitive to initial centroids (use k-means++) - Assumes spherical clusters of similar size - Requires pre-specifying k - Scale/normalize features before clustering - Run multiple times with different initializations

**K-Means++ Initialization**

**Algorithm:** Better initialization than random
1. Choose first centroid randomly 2. For each data point, compute distance D(x) to nearest centroid
3. Choose next centroid with probability $\propto$ D(x)$^2$ 4. Repeat until k centroids chosen
**Advantage:** Faster convergence, better final clusters

**Hierarchical Clustering**

**Linkage Methods:**
**Single Linkage:**

$$d(A,B) = \min_{a \in A, b \in B} d(a,b)$$

**Complete Linkage:**

$$d(A,B) = \max_{a \in A, b \in B} d(a,b)$$

**Average Linkage:**

$$d(A,B) = \frac{1}{|A||B|} \sum_{a \in A} \sum_{b \in B} d(a,b)$$

**Ward's Method:** Minimizes within-cluster variance

$$d(A,B) = \frac{|A||B|}{|A| + |B|} \|\mu_A - \mu_B\|^2$$

**Dendrogram:** Tree diagram showing cluster hierarchy
**When to use:** Unknown number of clusters, visualizing relationships

**Gap Statistic for Choosing k**

**Formula:**

$$Gap(k) = E[\log(W_k)] - \log(W_k)$$

where: - W$_k = within-clustersumofsquaresforkclusters - E[log(W_k)] = expectedvalueundernullreferencedistribution$
**Optimal k:** Choose smallest k where

$$Gap(k) \geq Gap(k+1) - s_{k+1}$$

where s$_{k+1} isstandarddeviation.$
**Interpretation:** Compares clustering to random data

## 9 Principal Component Analysis (PCA)

**PCA Overview - MEDIUM PROBABILITY (HIGH PRIORITY)**

**Goal:** Reduce dimensionality by finding directions of maximum variance.
**Steps:** 1. Standardize the data (mean 0, variance 1) 2. Compute covariance matrix 3. Calculate eigenvalues and eigenvectors 4. Sort eigenvectors by eigenvalues (descending) 5. Select top k eigenvectors (principal components) 6. Transform data to new space

## Covariance Matrix - HIGH PROBABILITY (HIGH PRIORITY)

**For standardized data X (n × p):**

$$\mathbf{C} = \frac{1}{n-1}\mathbf{X}^T\mathbf{X}$$

**Element (i,j):**

$$C_{ij} = \frac{1}{n-1}\sum_{k=1}^{n}(x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)$$

**Diagonal:** Variances of features
**Off-diagonal:** Covariances between features

## Eigenvalues and Eigenvectors - MEDIUM PROBABILITY (HIGH PRIORITY)

**Eigenvalue equation:**

$$\mathbf{C}\mathbf{v} = \lambda\mathbf{v}$$

where: - C = covariance matrix - v = eigenvector (principal component direction) - $\lambda$ = eigenvalue (variance along that direction)
**Properties:** - Each eigenvalue corresponds to one principal component - Larger eigenvalue = more variance explained - Eigenvectors are orthogonal (perpendicular)

## Variance Explained - HIGH PROBABILITY (HIGH PRIORITY)

**Total Variance:**

$$\text{Total Var} = \sum_{i=1}^{p}\lambda_i$$

**Variance Explained by PC i:**

$$\text{Var}_i = \frac{\lambda_i}{\sum_{j=1}^{p}\lambda_j}$$

**Cumulative Variance:**

$$\text{Cumulative Var}_k = \frac{\sum_{i=1}^{k}\lambda_i}{\sum_{j=1}^{p}\lambda_j}$$

**Rule of thumb:** Keep components explaining 80-95% of variance.

## PCA Transformation

**Transform data to PC space:**

$$\mathbf{Z} = \mathbf{X}\mathbf{W}$$

where: - X = original data (n × p) - W = matrix of k eigenvectors (p × k) - Z = transformed data (n × k)
**Reconstruction (approximate):**

$$\mathbf{X}_{\text{approx}} = \mathbf{Z}\mathbf{W}^T$$

## Quick Reference

**PCA Use Cases:** - Dimensionality reduction (visualization, speed) - Feature extraction - Noise filtering - Removing multicollinearity - Data compression
**Note:** PCA is sensitive to scaling - always standardize first!

## PCA Reconstruction Error

**Mean Squared Reconstruction Error:**

$$\text{MSE} = \frac{1}{n}\|\mathbf{X} - \mathbf{X}_{\text{approx}}\|_F^2$$

where $\|\cdot\|_F$ is Frobenius norm.
**Relation to Eigenvalues:**

$$\text{MSE} = \sum_{i=k+1}^{p} \lambda_i$$

**Interpretation:** Sum of discarded eigenvalues

## Kernel PCA - LOW PROBABILITY (HIGH PRIORITY)

**For non-linear relationships:**
Apply kernel trick to capture non-linear structure.
**Common Kernels:**
**RBF (Gaussian):**

$$K(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$

**Polynomial:**

$$K(x, y) = (x^T y + c)^d$$

**Sigmoid:**

$$K(x, y) = \tanh(\alpha x^T y + c)$$

**When to use:** Data with non-linear relationships, complex manifolds

## t-SNE (t-Distributed Stochastic Neighbor Embedding)

**Purpose:** Non-linear dimensionality reduction for visualization
**High-dimensional similarity:**

$$p_{j|i} = \frac{\exp(-\|x_i - x_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|x_i - x_k\|^2 / 2\sigma_i^2)}$$

**Low-dimensional similarity (t-distribution):**

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l}(1 + \|y_k - y_l\|^2)^{-1}}$$

**Cost Function (KL divergence):**

$$C = \sum_i KL(P_i \| Q_i) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

**When to use:** Visualizing high-dimensional data (2D/3D plots)
**Note:** Primarily for visualization, not for dimensionality reduction in modeling

## 10 Markov Chains

**Markov Chain Basics - MEDIUM PROBABILITY (HIGH PRIORITY)**

**Definition:** Stochastic process where future state depends only on current state (memoryless).
**Markov Property:**

$$P(X_{n+1} = j | X_n = i, X_{n-1}, \dots, X_0) = P(X_{n+1} = j | X_n = i)$$

**Transition Probability:**

$$P_{ij} = P(X_{n+1} = j | X_n = i)$$

**Transition Matrix - HIGH PROBABILITY (HIGH PRIORITY)**

**Matrix P:**

$$\mathbf{P} = \begin{pmatrix} P_{11} & P_{12} & \cdots & P_{1n} \\ P_{21} & P_{22} & \cdots & P_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ P_{n1} & P_{n2} & \cdots & P_{nn} \end{pmatrix}$$

**Properties:** - Row sums = 1: $\sum_{j=1}^{n} P_{ij} = 1$ - All entries $\geq 0$
**n-step Transition Probability:**

$$P_{ij}^{(n)} = (\mathbf{P}^n)_{ij}$$

**State Distribution after n steps:**

$$\boldsymbol{\pi}_n = \boldsymbol{\pi}_0 \mathbf{P}^n$$

**Stationary Distribution - MEDIUM PROBABILITY (HIGH PRIORITY)**

**Definition:** Distribution $\pi$ that doesn't change under transition

$$\boldsymbol{\pi} = \boldsymbol{\pi} \mathbf{P}$$

**Or equivalently:**

$$\boldsymbol{\pi}^T \mathbf{P} = \boldsymbol{\pi}^T$$

**With constraint:**

$$\sum_{i=1}^{n} \pi_i = 1$$

**Interpretation:** Long-run equilibrium probabilities.
**Note:** For ergodic chains, stationary distribution is unique and $\lim_{n\to\infty} \mathbf{P}^n$ converges to $\pi$.
**Computing Stationary Distribution:** Solve: $(\mathrm{P}^T - I)\pi = 0$ subject to $\Sigma\pi_i = 1$

### Markov Chain Properties

**Irreducibility:** Can reach any state from any other state
**Aperiodicity:** gcd of return times = 1
**Ergodicity:** Irreducible + aperiodic - Guarantees unique stationary distribution - Time-average = ensemble average
**Detailed Balance:**

$$\pi_i P_{ij} = \pi_j P_{ji}$$

If satisfied, $\pi$ is stationary distribution (reversible chain)
**Mean Recurrence Time:**

$$\mu_i = \frac{1}{\pi_i}$$

Average time to return to state i

### PageRank Algorithm

**Application of Markov Chains:** Google's PageRank
**Formula:**

$$PR(A) = (1-d) + d\sum_{i=1}^{n} \frac{PR(T_i)}{C(T_i)}$$

where: - d = damping factor (typically 0.85) - T$_i = pageslinkingtoA - C(T_i) = numberofoutboundlinksfromT_i$
**Matrix Form:**

$$\mathbf{PR} = (1-d)\mathbf{e} + d\mathbf{MPR}$$

**Interpretation:** Stationary distribution of random web surfer

## 11 Monte Carlo Methods

### Monte Carlo Integration - LOW PROBABILITY

**Estimate integral:**

$$\int_a^b f(x)\,dx \approx \frac{b-a}{n}\sum_{i=1}^{n} f(x_i)$$

where x$_i arerandomsamplesfromuniformdistributionon[a,b]$.
**Expected Value:**

$$\mathbb{E}[g(X)] \approx \frac{1}{n}\sum_{i=1}^{n} g(x_i)$$

where x$_i aresamplesfromdistributionofX$.

### Monte Carlo Simulation - MEDIUM PROBABILITY

**Steps:** 1. Define probability distributions for inputs 2. Generate random samples from distributions 3. Run deterministic computation for each sample 4. Aggregate results (mean, variance, percentiles)
**Law of Large Numbers:** As n $\to \infty$, sample mean converges to true mean.

**Standard Error:**

$$SE = \frac{s}{\sqrt{n}}$$

where s is sample standard deviation.
**Confidence Interval:**

$$\bar{x} \pm 1.96 \cdot \frac{s}{\sqrt{n}}$$

**Sample Size for Desired Accuracy:**

$$n = \left(\frac{z_{\alpha/2} \cdot s}{\epsilon}\right)^2$$

where $\varepsilon$ is desired margin of error.

**Importance Sampling**

**For rare events:** Sample from importance distribution
**Estimator:**

$$\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n}\frac{f(x_i)}{g(x_i)}h(x_i)$$

where: - f(x) = target distribution - g(x) = importance distribution (easier to sample) - h(x) = function of interest
**Variance Reduction:** Can significantly reduce MC variance
**When to use:** Rare events, tail probabilities, difficult distributions

## 12 Deterministic Models

**Deterministic vs Stochastic**

**Deterministic Model:** - No randomness - Same input always gives same output - Described by equations
**Examples:** - Linear equations: y = mx + b - Differential equations: dy/dt = f(t, y) - Physical laws: F = ma
**Stochastic Model:** - Includes randomness - Same input can give different outputs - Uses probability distributions
**Examples:** - Monte Carlo simulations - Markov chains - Random walk

**Common Deterministic Models**

**Linear Growth:**

$$y(t) = y_0 + rt$$

**Exponential Growth:**

$$y(t) = y_0 e^{rt}$$

**Logistic Growth:**

$$y(t) = \frac{K}{1 + \left(\frac{K-y_0}{y_0}\right)e^{-rt}}$$

where: - y$_0 = initialvalue - r = growthrate - K = carryingcapacity(forlogistic)$

### 12.1 Common Deterministic Models

**Linear Growth:**

$$y(t) = y_0 + rt$$

**Exponential Growth:**

$$y(t) = y_0 e^{rt}$$

**Logistic Growth:**

$$y(t) = \frac{K}{1 + \left(\frac{K-y_0}{y_0}\right) e^{-rt}}$$

where: - y$_0 = initialvalue - r = growthrate - K = carryingcapacity(forlogistic)$

## 13 Bias-Variance Tradeoff

**Quick Reference**

**Model Complexity:** - Simple models: High bias, low variance (linear regression) - Complex models: Low bias, high variance (high-degree polynomial) - Ensemble methods: Balance both (random forest, boosting)
**Regularization:** Increases bias, decreases variance

## 14 Feature Selection and Engineering

**Filter Methods**

**Correlation with Target:** Select features with highest —correlation— with target variable.
**Chi-Square Test:** For categorical features, test independence with target.
**Mutual Information:**

$$I(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$$

Measures dependency between features and target.
**Variance Threshold:** Remove features with low variance (nearly constant).

**Wrapper Methods**

**Forward Selection:** 1. Start with no features 2. Add feature that improves model most 3. Repeat until no improvement
**Backward Elimination:** 1. Start with all features 2. Remove feature that hurts least 3. Repeat until further removal hurts
**Recursive Feature Elimination (RFE):** Train model, rank features by importance, remove least important, repeat.
**When to use:** Better performance than filters, but computationally expensive.

**Embedded Methods**

**LASSO:** L1 penalty automatically selects features (sets coefficients to zero)
**Ridge:** L2 penalty shrinks but doesn't eliminate features
**Elastic Net:** Combination of L1 and L2
**Tree-Based Feature Importance:** Based on information gain or impurity decrease.
**Advantage:** Feature selection during model training.

## 15 Ensemble Methods

**Bagging (Bootstrap Aggregating) - MEDIUM PROBABILITY (HIGH PRIORITY)**

**Algorithm:** 1. Create B bootstrap samples (sampling with replacement) 2. Train model on each sample 3. Aggregate predictions (average or vote)
**Purpose:** Reduce variance, prevent overfitting
**Examples:** Random Forest, Bagged Decision Trees
**Prediction:**

$$\hat{y} = \frac{1}{B}\sum_{b=1}^{B}\hat{f}_b(x)$$

**When to use:** High-variance models (decision trees, neural networks)

**Boosting - MEDIUM PROBABILITY (HIGH PRIORITY)**

**Idea:** Sequentially train models, each correcting previous errors
**AdaBoost Weight Update:**

$$w_i^{(t+1)} = w_i^{(t)} \exp(\alpha_t \mathbb{I}(y_i \neq \hat{y}_i))$$

where:

$$\alpha_t = \frac{1}{2}\log\frac{1 - varepsilon_t}{varepsilon_t}$$

$\varepsilon_t = weightederrorrate$
**Gradient Boosting:** Fit new model to residuals (negative gradient) of previous model.
**Purpose:** Reduce bias, improve accuracy
**Examples:** AdaBoost, XGBoost, LightGBM, CatBoost
**When to use:** Need high accuracy, can handle complex patterns

**Stacking (Stacked Generalization)**

**Two-Level Approach:**
**Level 0 (Base Models):** Train multiple diverse models on training data.
**Level 1 (Meta-Model):** Train on predictions from base models.
**Prediction:**

$$\hat{y} = g(f_1(x), f_2(x), \ldots, f_M(x))$$

where g is meta-model, f$_i arebasemodels$.
**When to use:** Combining different model types for best performance.

# 16 Time Series Analysis (Brief)

## Moving Average (MA)

**Simple Moving Average:**

$$MA_t = \frac{1}{k}\sum_{i=0}^{k-1} y_{t-i}$$

**Weighted Moving Average:**

$$WMA_t = \sum_{i=0}^{k-1} w_i y_{t-i}$$

**Exponential Smoothing:**

$$S_t = \alpha y_t + (1-\alpha)S_{t-1}$$

where 0 ¡ $\alpha$ ¡ 1 is smoothing parameter.
**When to use:** Smoothing data, forecasting, trend detection.

## Autocorrelation

**Autocorrelation Function (ACF):**

$$\rho_k = \frac{\text{Cov}(y_t, y_{t-k})}{\text{Var}(y_t)}$$

Correlation between series and itself at lag k.
**Partial Autocorrelation (PACF):** Correlation between y$_tandy_{t-k}afterremovingeffectofintermediatelags.$
**When to use:** Identifying AR/MA orders, checking randomness.

# 17 Quick Reference Tables

## 17.1 Distribution Properties

| Distribution | E[X] | Var(X) | Use Case |
|---|---|---|---|
| Bernoulli(p) | p | p(1-p) | Single trial |
| Binomial(n,p) | np | np(1-p) | n trials |
| Poisson($\lambda$) | $\lambda$ | $\lambda$ | Rare events |
| Normal($\mu$,$\sigma^2$) | $\mu$ | $\sigma^2$ | Continuous, symmetric |

## 17.2 Test Selection Guide

| Scenario | Test |
|---|---|
| One sample, known $\sigma$ | Z-test |
| One sample, unknown $\sigma$ | t-test |
| Two samples, independent | Two-sample t-test |
| Two samples, paired | Paired t-test |
| Proportions | Z-test for proportions |

### 17.3 Common Z-scores

| Confidence Level | $\alpha$ | Z-score |
|---|---|---|
| 90% | 0.10 | 1.645 |
| 95% | 0.05 | 1.96 |
| 99% | 0.01 | 2.576 |
| 99.9% | 0.001 | 3.291 |

### 17.4 Effect Size Interpretations

| Measure | Small | Medium | Large |
|---|---|---|---|
| Cohen's d | 0.2 | 0.5 | 0.8 |
| Pearson r | 0.1 | 0.3 | 0.5 |
| $R^2$ | 0.01 | 0.09 | 0.25 |
| Cramér's V | 0.1 | 0.3 | 0.5 |
| Eta-squared ($\eta^2$) | 0.01 | 0.06 | 0.14 |

### 17.5 Distance Metrics Comparison

| Distance | Formula | Use Case |
|---|---|---|
| Euclidean | $\sqrt{\sum(x_i - y_i)^2}$ | General purpose, KNN |
| Manhattan | $\sum \lvert x_i - y_i \rvert$ | Grid-like paths, robust |
| Cosine | $1 - \frac{x \cdot y}{\lVert x \rVert \lVert y \rVert}$ | Text, high dimensions |
| Minkowski (p=3) | $(\sum \lvert x_i - y_i \rvert^p)^{1/p}$ | Generalization |
| Hamming | Count of differences | Binary/categorical |

### 17.6 Model Complexity and Regularization

| Method | Bias | Variance | Sparsity |
|---|---|---|---|
| OLS | Low | High | No |
| Ridge (L2) | Medium | Medium | No |
| LASSO (L1) | Medium | Medium | Yes |
| Elastic Net | Medium | Medium | Yes |

### 17.7 Evaluation Metrics by Task

| Task | Recommended Metrics |
|---|---|
| Regression | RMSE, MAE, $R^2$, Adjusted $R^2$ |
| Binary Classification | Accuracy, Precision, Recall, F1, AUC-ROC |
| Multi-class Classification | Accuracy, Macro/Micro F1, Confusion Matrix |
| Imbalanced Classification | F1-Score, Precision-Recall AUC, MCC |
| Clustering | Silhouette Score, Davies-Bouldin Index, WCSS |
| Ranking | NDCG, MAP, MRR |

### 17.8 Algorithm Comparison

| Algorithm | Type | Pros | Cons |
|---|---|---|---|
| KNN | Supervised | Simple, no training | Slow prediction, memory |
| K-Means | Unsupervised | Fast, scalable | Need k, spherical clusters |
| Naive Bayes | Supervised | Fast, works with small data | Independence assumption |
| Decision Tree | Supervised | Interpretable | Overfitting |

## 18 Additional Important Distributions

**T-Distribution - VERY HIGH PROBABILITY (HIGH PRIORITY)**

**Probability Density Function:**

$$f(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2}, \quad t \in \mathbb{R}$$

where $\nu$ = degrees of freedom, $\Gamma(\cdot)$ = Gamma function

**Properties:**

- $\mathbb{E}[T] = 0$ for $\nu > 1$
- $\text{Var}(T) = \frac{\nu}{\nu-2}$ for $\nu > 2$
- As $\nu \to \infty$, $t_\nu \to \mathcal{N}(0, 1)$
- Heavier tails than normal (more robust to outliers)
- Symmetric about 0

**Common Critical Values:** $t_{0.025,10} = 2.228$, $t_{0.025,20} = 2.086$, $t_{0.025,30} = 2.042$, $t_{0.025,\infty} = 1.96$

**Chi-Square Distribution $\chi^2(\nu)$ - HIGH PROBABILITY (HIGH PRIORITY)**

**Probability Density Function:**

$$f(x) = \frac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{(\nu/2)-1} e^{-x/2}, \quad x > 0$$

**Properties:** $\mathbb{E}[\chi^2] = \nu$, $\text{Var}(\chi^2) = 2\nu$

If $Z_1, \ldots, Z_\nu \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$, then $\sum_{i=1}^{\nu} Z_i^2 \sim \chi^2(\nu)$

**Variance Estimation:** $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$

**F-Distribution $F(\nu_1$**

If $U \sim \chi^2(\nu_1)$ and $V \sim \chi^2(\nu_2)$ independent:

$$F = \frac{U/\nu_1}{V/\nu_2} \sim F(\nu_1, \nu_2)$$

**Properties:** $\mathbb{E}[F] = \frac{\nu_2}{\nu_2-2}$ for $\nu_2 > 2$

**Comparing Variances:** $\frac{s_1^2}{s_2^2} \sim F(n_1 - 1, n_2 - 1)$

**Gamma Distribution $\Gamma(\alpha, \beta)$ - MEDIUM PROBABILITY (HIGH PRIORITY)**

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0$$

**Parameters:** $\alpha$ = shape, $\beta$ = rate

**Properties:** $\mathbb{E}[X] = \frac{\alpha}{\beta}$, $\text{Var}(X) = \frac{\alpha}{\beta^2}$

**Special Cases:** $\Gamma(1, \beta) = \text{Exp}(\beta)$, $\Gamma(\nu/2, 1/2) = \chi^2(\nu)$

## 19 Sampling Distributions and Convergence

**Sampling Distribution of $\bar{X}$ - VERY HIGH PROBABILITY (HIGH PRIORITY)**

For i.i.d. $X_1, \dots, X_n$ with $\mathbb{E}[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2$:
**Mean and Variance:**

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad \text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

**Distribution:**

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{(if } X_i \text{ normal or } n \text{ large by CLT)}$$

**Standardized:**

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1) \quad \text{or} \quad T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

**Chi-Square for Variance - HIGH PROBABILITY**

For $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$:

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$$

**CI for $\sigma^2$:**

$$\left[\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}\right]$$

## 20 Non-Parametric Methods

**Kernel Density Estimation (KDE)**

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

where $K(\cdot)$ = kernel, $h$ = bandwidth
**Gaussian Kernel:** $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$
**Silverman's Bandwidth:** $h = 0.9 \cdot \min\left\{s, \frac{\text{IQR}}{1.34}\right\} \cdot n^{-1/5}$

**Bootstrap Resampling - MEDIUM PROBABILITY (HIGH PRIORITY)**

1. Draw $B$ samples of size $n$ with replacement 2. Calculate $\hat{\theta}^{(b)}$ for each sample 3. Estimate SE and CI
**Bootstrap SE:**

$$\text{SE}_{\text{boot}} = \sqrt{\frac{1}{B-1} \sum_{b=1}^{B} \left(\hat{\theta}^{(b)} - \bar{\theta}^*\right)^2}$$

**Percentile CI:** $[\hat{\theta}_{(0.025)}, \hat{\theta}_{(0.975)}]$

## 21 High-Dimensional Statistics

**Curse of Dimensionality - MEDIUM PROBABILITY (HIGH PRIORITY)**

**Volume of Hypersphere:**

$$V_d(r) = \frac{\pi^{d/2}}{\Gamma(d/2+1)} r^d$$

**Key Issues:**

- Distance concentration: $\frac{d_{\max} - d_{\min}}{d_{\min}} \to 0$ as $d \to \infty$
- Most volume near surface
- Data becomes sparse
- $n$ required grows exponentially with $d$

**Mitigation:** - Dimensionality reduction (PCA) - Feature selection - Regularization - Distance metric learning

## 22 Information Theory for ML

**Entropy - HIGH PROBABILITY (HIGH PRIORITY)**

**Shannon Entropy (Binary):**

$$H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$$

**General Entropy:**

$$H(X) = -\sum_{i=1}^{k} p_i \log_2 p_i = \mathbb{E}[-\log_2 P(X)]$$

**Properties:**

- $H(X) \geq 0$ (non-negative)
- $H(X) = 0$ iff distribution is deterministic
- Maximum when uniform: $H_{\max} = \log_2 k$
- Concave function

**Gini Impurity - HIGH PROBABILITY (HIGH PRIORITY)**

$$G(p) = 2p(1-p) = 1 - \sum_{i=1}^{k} p_i^2$$

**Relationship:** $H(p) \geq G(p)$ for all $p$

**For Decision Trees:** - Information Gain: $\text{IG} = H(S) - \sum_v \frac{|S_v|}{|S|} H(S_v)$ - Gini Gain: $\Delta G = G(S) - \sum_v \frac{|S_v|}{|S|} G(S_v)$

**Cross-Entropy Loss**

**Binary Classification:**

$$L = -\frac{1}{n}\sum_{i=1}^{n}[y_i \log(\hat{p}_i) + (1-y_i)\log(1-\hat{p}_i)]$$

**Multi-Class:**

$$L = -\frac{1}{n}\sum_{i=1}^{n}\sum_{c=1}^{C} y_{ic}\log(\hat{p}_{ic})$$

**Interpretation:** Measure of difference between two probability distributions

## 23 Advanced Regression Concepts

**Residual Analysis - HIGH PROBABILITY (HIGH PRIORITY)**

**Residual:** $e_i = y_i - \hat{y}_i$
**Standardized Residual:**

$$r_i = \frac{e_i}{\hat{\sigma}\sqrt{1-h_{ii}}}$$

where $h_{ii} = [\mathbf{H}]_{ii}$ (leverage)
**Studentized Residual:**

$$t_i = \frac{e_i}{\hat{\sigma}_{(i)}\sqrt{1-h_{ii}}} \sim t_{n-p-2}$$

**Cook's Distance (Influence):**

$$D_i = \frac{(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)})^T(\mathbf{X}^T\mathbf{X})(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)})}{p\hat{\sigma}^2} = \frac{r_i^2}{p}\frac{h_{ii}}{1-h_{ii}}$$

**Rule:** $D_i > \frac{4}{n}$ or $D_i > 1$ suggests influential point

**Variance Inflation Factor (VIF)**

For predictor $X_j$:

$$VIF_j = \frac{1}{1-R_j^2}$$

where $R_j^2$ = R-squared from regressing $X_j$ on other predictors
**Interpretation:** - $VIF < 5$: No multicollinearity - $5 \leq VIF < 10$: Moderate - $VIF \geq 10$: Severe multicollinearity

**Akaike Information Criterion (AIC)**

$$AIC = 2k - 2\ln(\hat{L})$$

where $k$ = number of parameters, $\hat{L}$ = maximized likelihood

**For Linear Regression:**

$$AIC = n \ln \left( \frac{SSE}{n} \right) + 2k$$

**Bayesian IC (BIC):**

$$BIC = k \ln(n) - 2 \ln(\hat{L})$$

**Rule:** Lower AIC/BIC indicates better model (balances fit vs complexity)

## 24 Cross-Validation and Model Selection

**K-Fold Cross-Validation - HIGH PROBABILITY (HIGH PRIORITY)**

**Procedure:** 1. Split data into $K$ folds 2. For each fold $k = 1, \ldots, K$: - Train on $K - 1$ folds - Validate on fold $k$ - Compute error $E_k$ 3. Average: $CV_{(K)} = \frac{1}{K} \sum_{k=1}^{K} E_k$
**Special Cases:** - $K = n$: Leave-One-Out CV (LOOCV) - $K = 5$ or $K = 10$: Common choices
**Bias-Variance:** - Small $K \to$ higher bias, lower variance - Large $K \to$ lower bias, higher variance

**Train-Test Split**

**Common Splits:** - 70/30 train/test - 80/20 train/test - 60/20/20 train/validation/test
**Stratified Sampling:** Preserve class proportions in splits

**— END OF FORMULA SHEET —**

*Good luck on your exam!*